

PT Message Generation – Production Design Document

Problem / Motivation

Problem:
Physical Therapists (PTs) are expected to provide empathetic, personalized feedback to patients after each recovery session. While this practice supports patient motivation and engagement, it is repetitive, time-consuming, and inconsistent across practitioners.

Motivation:
Automating this task offers an opportunity to:

- Reduce PT workload
- Deliver timely and consistent communication
- Ensure every patient receives thoughtful encouragement

To succeed, the system must be clinically safe, customizable, and indistinguishable from a human-generated message — never robotic, generic, or AI-sounding.

Overview of the Solution

We propose a modular, LLM-powered system that generates short (2–4 sentence) post-session messages using structured data. The messages will be reviewed by PTs through a dedicated UI before being delivered to patients.

Core Components

1. **Feature Transformer** – Converts structured session metrics into a natural-language summary.
 2. **Prompt Constructor** – Builds LLM prompts with explicit tone, personalization, and safety instructions.
 3. **Message Generator (LLM)** – Uses GPT-4 or OSS LLMs (e.g., Mistral) to produce human-like messages.
 4. **PT Review Interface** – A user-friendly UI allowing PTs to accept, edit, or reject messages.
 5. **Feedback Loop** – Logs therapist actions to guide future refinement or fine-tuning.
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Success Metrics

Category	Metric	Goal
Message Quality	% accepted without edits	≥ 80%
Engagement	% of sessions with reviewed messages	≥ 90%
Efficiency	Avg. time saved per PT per session	≥ 30 seconds
Personalization	% of messages with session-specific content	≥ 95%

Category	Metric	Goal
Safety	% of messages with flagged clinical inaccuracies	≤ 0.5%
Feedback Integration	Rejection reason frequency reduction	Downward trend

Methodology

Data Inputs

- Structured features per session (e.g., pain, fatigue, exercises skipped)
- No labeled training data initially
- PT feedback (accept/edit/reject) used for iterative evaluation

Model Strategy

- **Initial Phase:** Use GPT-4 with few-shot prompting
- **Fallback:** Template-based messages for incomplete outputs
- **Future State:** Fine-tune OSS model on curated PT-reviewed samples

Clinical and Technical Safeguards

- Hardcoded tone and structural constraints in prompts
- Explicit avoidance of hallucinations and medical advice
- All messages routed through human-in-the-loop validation

Deployment Architecture

Layer	Tooling	Purpose
LLM API	OpenAI GPT-4 / OSS (Mistral, etc.)	Message generation
Prompt Engine	Python	Prompt generation from session data
API Backend	FastAPI	Message orchestration, feedback logging
Frontend	Streamlit (PoC/MVP) → React	Therapist-facing message review interface
Scheduling	Airflow	Message batch generation and feedback monitoring
Streaming (opt.)	Kafka / Redis Streams	Real-time session ingestion (optional at MVP stage)
Storage	PostgreSQL	Stores sessions, prompts, messages, feedback logs
Monitoring	Prometheus + Grafana	Token usage, latency, completion success tracking
Model Tracking	MLflow / Weights & Biases	Prompt versioning and experimentation tracking

Layer	Tooling	Purpose
Infrastructure	Docker + Kubernetes	Containerized, auto-scalable production deployment

Delivery Phases & Timeline

Phase	Duration	Key Deliverables
Phase 1 – PoC	1.5 months	Manual prompt testing, sample messages, internal validation
Phase 2 – MVP	4–6 months	Prompt engine, GPT-4 integration, Streamlit UI, feedback loop
Phase 3 – Productionization	6–9 months	Full stack build, monitoring, React UI, Kubernetes deployment
Phase 4 – Continuous Improvement	Ongoing	Fine-tuning, RAG, drift detection, multilingual support

Total estimated time to production maturity: 12–16 months

Timeline Risks and Mitigation

Risk	Mitigation Strategy
Staff turnover	Modular architecture, written documentation, onboarding guides
Lack of LLM expertise	Start with prompt-based PoC to defer fine-tuning complexity
Feedback delays	Incentivize early PT participation, use internal QA proxy
Budget & pricing constraints	Track token usage; switch to OSS models or batching as needed
Data annotation gaps	Leverage weak supervision; mine accepted messages for tuning
Compliance review bottlenecks	Involve clinical experts early, automate tone/safety checks

Message Generation Pipeline

1. Data Ingestion
- Sessions arrive via API or stream; stored in PostgreSQL
2. Prompt Generation
- Prompt builder assembles structured user prompt
3. LLM Call
- LLM generates message with latency/token metadata captured

4. PT Review

- Message is presented in UI with accept/edit/reject tagging

5. Feedback Storage

- Logged into DB and optionally fed into a fine-tuning dataset

Future Improvements

- **Fine-tuning OSS models** to reduce cost and increase control
- **Multilingual support** for patients across regions
- **Configurable tone/length** per therapist or patient type
- **Session trend tracking** to reference previous progress
- **Retrieval-Augmented Generation (RAG)** to improve variety
- **Safety classifier** for hallucination or tone mismatches
- **Reinforcement via PT feedback** to gradually improve message quality

Conclusion

This system is designed to balance clinical safety, PT usability, and technical feasibility while keeping costs and complexity in check. By deploying in progressive phases, we can derisk core components, gather feedback early, and scale toward a robust human-in-the-loop assistant that delivers high-value, human-like patient encouragement at scale.